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MILITARY SPECIFICATION

RECORDING SET, SIGNAL DATA A/A24U-6

1. SCOPE

1.1 This specification covers the requirements for an airborne signal data recording set, designated A/A24U-6. The recording set measures flight parameters relating to aircraft structural loads and encodes these parameters in digital form for recording on magnetic tape. The digital information on the magnetic tape shall be either in an IBM-compatible format or can be made IBM compatible through use of ground conversion equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture- and Fungus-resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-P-22808	Paint, Epoxy, Hydraulic Fluid Resistant
MIL-P-23377	Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant
MIL-C-26482	Connector, Electric, Circular, Miniature, Quick Disconnect
MIL-C-26500	Connector, General Purpose, Electrical Miniature, Circular, Environment Resisting, 200°C Ambient Temperature
MIL-C-26518	Connector, Electrical, Miniature, Rack and Panel, Environment Resisting, 200°C Ambient Temperature

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-415	Test Points and Test Facilities Design Standard for
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

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MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MS24385	Fitting End, Flared Tube Connection and Gasket Seal Precision Type, Standard Dimensions
MS25271	Relay, 10 Amp, 4 PDT, Class B8, Sealed, with Solder Mounts
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The recording set shall consist of the following major components:

a. MXK-316/A24U-6 recorder mechanism assembly

b. MTK-43/A24U-6 mount, resilient

c. MXK-315/A24U-6 magazine, recorder

d. Tape transport case.

3.2.1 Input filters shall be required to meet response and other dynamic characteristics specified herein and may be an integral part of the particular sensing unit.

3.3 General specification. The requirements of MIL-E-5400 as specified for class 2 equipment shall apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.4 Materials

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. When used and not hermetically sealed, they shall be treated with a fungicidal agent conforming to MIL-T-152. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to atmospheric and climatic conditions to which the transducer will be subjected during storage or normal services.

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3.4.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.4 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts, except where magnetic materials are essential.

3.4.5 Nonferrous materials. Nonferrous materials shall be used for all parts, except where ferrous materials are essential.

3.4.5.1 Protection of nonferrous materials. Nonferrous materials contained within hermetically sealed enclosures shall be considered suitably protected from corrosion. Requirements specified for fungicidal and corrosion protective treatment and anodizing of aluminum alloy parts will not be applicable for parts within hermetically sealed enclosures.

3.4.6 Toxic and corrosive fumes. Materials used in the construction of the recording set shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.4.7 Protective treatment. When materials are used in the construction of the recording set that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.8 Materials shall be of the highest quality and the lightest weight suitable for the purpose intended.

3.4.9 Finishes. Interior surfaces shall be finished in accordance with MIL-E-5400. Exterior finishes for equipment bays and components shall be in accordance with the following:

a. Cleaning and surface treatment in accordance with the applicable requirements of MIL-E-5400.

b. One coat of epoxy primer conforming to MIL-P-23377.

c. One coat of gray epoxy paint conforming to MIL-P-22808, color No. 36231.

3.4.10 Lubrication. During assembly, the recording set shall be lubricated, as necessary, and shall require a minimum of lubrication throughout its normal service life. Lubrication service shall be limited to the major servicing time specified in 3.5.10.1. Operation of the recording set in any position shall not affect lubrication.

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3.5 Design and construction. The recording set shall be designed to measure and record up to 24 input parameters at a total system sampling rate of 240 samples per second. The total samples per second, information bandwidth (Hz), filtering measurements (db/octave), and electrical characteristics of each input channel shall be specified by the procuring activity. The nominal sampling rate of any channel shall be five times the maximum value of frequency response (bandwidth) desired by the procuring activity (i.e., bandwidth 0-6 Hz would require 30 samples per second). All recorder components, such as submultiplexers, signal conditioners, filters, et cetera, necessary to achieve the 24-channel capability shall be included in the recorder mechanism assembly. The recording set shall measure and convert analog signals into digital form for recording, and shall be as simple and foolproof as possible in operation, installation, and maintenance. The recording medium and record/playback heads shall be contained in a detachable magazine assembly suitable for easy handling. The magazine assembly shall be attachable directly to the recorder mechanism for data recording and to the necessary ground handling equipment for processing. The recording set shall be so designed and constructed that no fixed part or assembly will become loose, no moving or movable part or control will be shifted in setting, position, or adjustment, and no degradation will result in performance below that specified herein during operation or after storage in the specified ambient conditions. All recorder mechanisms and magazine assemblies and all major components of the recording set furnished under this specification shall be common in design and interchangeable. As applicable, separate adjustments to the playback unit for any differences in individual recorder performance shall not be permitted.

3.5.1 Maintenance. The recording set shall be designed to facilitate, as much as possible, disassembly, repair or overhaul, service maintenance, and reassembly by tools and items of maintenance normally available commercially.

3.5.2 Recording time. The recording set shall be capable of measuring and recording data on all data channels for at least 25 continuous hours without changing magazines. Variations of average tape speed among recording sets shall be within ±2 percent of the specified record speeds.

3.5.3 Recording method. The recording set shall sequentially sample the analog input signals. Each shall be encoded into a binary form (consistant with system accuracy requirements specified herein) along with pulses for clocking and synchronization if necessary to conform to IBM format. This data shall be recorded on the magnetic tape.

3.5.4 Electrical power. Electrical power available for the recording set shall be nominal 28V dc and nominal 115V 400 Hz single-phase ac. The total power requirements for the airborne recording set shall be specified by the manufacturer and shall not exceed 100w with all components functioning under maximum load conditions. The d-c power requirements shall not exceed 10w.

3.5.4.1 Equipment. The equipment shall meet all applicable requirements of MIL-STD-704 and shall operate from the power sources defined therein equivalent to

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category B, including the emergency conditions, except degraded performance will be permitted in the emergency conditions with the approval of the procuring activity.

3.5.5 Connectors and wiring. Connectors used in the recording set shall conform to MIL-C-26500 or MIL-C-26518 except that connectors to interface the set may be in accordance with MIL-C-26482. Electrical receptacles (except blind mated) shall be easily accessible for maintenance when the recording set is installed in the aircraft and shall be covered with a protective cap attached to the recording set by a retaining chain. Arrangement of the wiring shall be as specified by the procuring activity.

3.5.5.1 Test points. A single receptacle shall be provided on the face of the recording set in which voltages shall be available to measure proper functioning of the system. Test points for a shop-level test set shall include motor current, plus and minus voltage supplies, A-D converter output, timer, ramp trigger system oscillator output, multiplexer output, and any other test signals needed to check out the system and isolate by functional component any trouble area. All adjustments shall be available from the face of the recording set. Adjustments shall not be required in less than 500 hours of operation. The receptacle shall be covered by a protective screw-on cap which shall be attached to the recording set by a retaining chain. The recording set shall have the built-in test equipment (BITE) necessary to isolate failure down to the line replaceable unit (LRU) level testing.

3.5.6 Remote switch. Recording shall be controlled by a remotely located ON-OFF switch. Power shall be connected directly to electronic elements, heating elements, and transducers for warmup provisions.

3.5.7 Mounting. The recording set shall operate properly when mounted in any position. An external vibration isolator assembly, including mounting plate, shall be provided, if required, with each recording set. Provisions shall be made for easily mounting the recording set to the mounting plate in the aircraft. Each transducer shall have its own mounting bracket, as applicable.

3.5.8 Calibration. After application of aircraft power for 3 minutes, closure of the remote switch shall generate and record on tape the following calibration sequence. The tape shall not be running prior to actuation of the remote switch. The signals shall be recorded on all channels simultaneously. In each case, this procedure shall be required prior to recording of data.

Full scale for 5 ±1/2 seconds

Half scale for 5 ±1/2 seconds

Zero for 5 ±1/2 seconds

3.5.9 Reliability program. Except as specified herein, a reliability assurance program in accordance with MIL-STD-785 shall be established, maintained, and documented by the contractor.

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3.5.9.1 Reliability. The recorder shall have a minimum mean-time-between-failures (MTBF) of 400 hours which shall be demonstrated as specified in 4.6.21.

3.5.9.2 Longevity. The recorder shall have a minimum total operating life (longevity) of 10,000 hours with reasonable servicing and replacement of parts. All servicing and all parts inherently requiring replacement in less than 10,000 hours must be called out in the technical proposal along with their individual estimated service life.

3.5.10 Maintainability program. The maintainability program shall be in accordance with MIL-STD-470. The contractor shall submit, as a part of the technical proposal, an analytical maintenance plan which shall include, but not be limited to, defining levels of support required (line shop depot) to maintain the recording set, expected failure modes, and equipment required to repair any failed component. An estimate of the maintenance manhours required to support the recording set per 100 hours of operation shall also be provided.

3.5.10.1 Maintenance and repair cycles. The recording set shall have a mean maintenance down time ($\overline{M}$) of not more than 30 minutes. There shall be no scheduled maintenance (including maintenance inspection and parts replacement) required except as permitted in 3.5.10.

3.5.10.2 The recording set shall operate as specified herein without major servicing for a period of 500 hours of continuous operation. Major servicing shall be considered as servicing of any nature which requires removal of the recording set from the aircraft or which requires more than 45 minutes of actual effort on the recording set.

3.5.10.3 Service and access. The equipment shall be so designed that one man can accomplish any single maintenance action. Equipment design shall include provisions for safe and convenient electromechanical access to facilitate maintenance at all levels. Relative failure rates shall be a prime consideration in arranging equipment for maintenance access.

3.5.10.4 Simplicity. Simple devices are preferable to complex devices. All functional parts of the LRU's shall be contained in separately removable (plug-in) modules except for interconnecting wiring, electrical connectors, and LRU structure and mounting provisions, unless an exception is justified by the manufacturer and approved by the procuring activity. To the maximum extent possible, in line with the operation and accuracy requirements of this specification, like components, transistors, diodes, and other parts shall be used to reduce logistics.

3.5.10.5 Organizational level maintenance. The organizational level maintenance shall be conducted using the BITE or the AGE. The BITE shall test the recorder/multiplexer from input to recording heads, and shall also test for tape transport operation. The AGE shall be used in conjunction with the recorder/multiplexer to isolate a defective input channel. Organizational level maintenance shall be performed by 3- and 5-level enlisted personnel. Any tasks requiring 7-level personnel shall be justified. In no case will the utilization of 7-level personnel in active maintenance

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repair exceed one-fourth the permissible utilization of 5-level technicians. The design emphasis shall be directed to on-aircraft maintenance environmental conditions in any climate from -54° to +71°C, by personnel wearing clothing such as heavy gloves, and in rain, snow, blowing sand and dust, et cetera, without any cleanup being required except normal wiping up.

3.5.10.6 Intermediate level maintenance. Intermediate level maintenance shall consist of isolation of the defective assembly, subassembly, or piece part, and replacement or repair of the defective item. Intermediate shop level maintenance actions shall be performed by 3-, 5-, and 7-level personnel. Any maintenance tasks requiring 7-level personnel shall be justified. In no case will the utilization of 7-level personnel in the active maintenance repair exceed one-fourth the permissible utilization of 5-level technicians. Intermediate level maintenance shall be capable of being performed over the range of ambient temperatures from 0° to +55°C. Test provisions with the AGE shall provide a 0.99 probability of identifying the malfunctioning part or assembly and of substantiating satisfactory completion of repairs. Test points shall be selected in accordance with MIL-STD-415.

3.5.10.7 Base self-sufficiency. All repairable LRU's, modules, assemblies, subassemblies, and components shall be considered for repair at base levels by replacement of nonrepairable bits or pieces. The following are minimum requirements:

a. A minimum of 40 percent of the total number of subassemblies used to repair reparable LRU's shall be repairable at the base level. This limitation may be waived if the contractor substantiates with MTBR's sufficiently high, i.e.,4000 hours MTBR or higher.

3.6 Performance. The recorder shall be capable of operating as specified herein under the following conditions:

a. Applied acceleration of 20g along any axis

b. Operation at temperatures ranging from -54° to +71°C at a maximum altitude of 80,000 feet

c. Vibration of 10g's at room temperature, -30°, and +55°C

d. Shock of 20g's for 10 milliseconds along any axis (nonoperating)

e. Humidity up to 95 percent

f. Salt fog for a minimum period of 48 hours

g. Fungus as encountered in tropical climates.

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3.6.1 Error percentages. Unless otherwise specified, all error percentages shall be percentages of full-scale output. Errors specified for the recording set shall not include errors attributable to the transducers unless noted otherwise. Testing required to demonstrate adherence to performance shall be performed using the analog output signals from the signal data converter.

3.6.2 System accuracy. The combined error of the signal data recorder (excluding sensors) and the signal data converter shall be determined using programmable signal generator as the input signal and using appropriate monitoring and recording test equipment. The combined error (system accuracy) of the system shall include individual or simultaneous interacting parameters such as nonlinearity, noise, drift, frequency response, distortion, environmental effects, and any causes of error attributable to the recorder. The recorder accuracy shall meet the following error distribution which shall not apply to any other performance requirements specified herein.

Percent of All Test Measurements	Percent Error
90	±2.0 or less
95	±2.5 or less
98	±3.0 or less

The remaining 2 percent of all test measurements shall follow the relation:

$$\frac{1}{\sqrt{2\pi}} \int_{\frac{-E}{0.0168}}^{\frac{E}{0.0168}} e^{-\frac{x^2}{2}}\,dx$$

Where e = base natural log
E = Percent error

3.6.3 Linearity. Nonlinearity errors shall be expressed as deviations from a straight line drawn between the ideal end points of a calibration graph of input-output values. Data for this graph shall be obtained as specified in 4.6.3. Nonlinearity errors existing between the actual input value of any recorded parameter and the analog playback voltage from the signal data converter shall not exceed ±2.0 percent.

3.6.4 Noise (rms). With static input signals applied to the recorder, the ratio of full scale signal to rms noise at the signal data converter output shall be 40 db or greater.

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3.6.5 Noise (peak-to-peak). With static input signals applied to the recorder, the following distribution upon playback of any channel shall be met during 10 minutes of recording. The pulses shall include noise from any source or cause.

Peak-to-Peak amplitude of Noise Pulses	Number of Pulses Per Channel Allowed
LSB (Least significant bit)	8
2nd LSB	4
3rd LSB	2
4th LSB	1
5th LSB	1
6th LSB	1
7th LSB (as applicable)	1

3.6.6 Drift and stability. The instability of the output analog voltage from the signal data converter for a constant input signal to the recorder shall not exceed ±1.0 percent. This parameter and the required testing shall include effects of 1-bit resolution.

3.6.7 Crosstalk. Any form of crosstalk shall be included in the measurement and performance requirements of 3.6.4 and 3.6.5.

3.6.8 Dropouts and dropins. Any form of dropouts and their affect on system performance shall be included in the measurement and performance requirements of 3.6.4 and 3.6.5. Dropouts shall include, by definition, such phenomena as digital bit loss from any cause, dropins, and digital jitter.

3.6.9 Frequency response. The recording set shall record input signals in the frequency range of 0 through 6 Hz or 0 through 12 Hz, as applicable. When measured as an analog voltage, the frequency response shall be flat within ±0.2 db of static response through the frequency ranges specified herein.

3.6.10 Temperature sensitivity. The recorder shall operate properly at all ambient temperatures from -54° to +71°C and shall meet all performance requirements specified herein when operating in this temperature range, except degradation of 1.0 percent from room ambient temperature shall be allowed when operating in the range from -54° to -30°C and in the range from +55° to +71°C. This allowable degradation shall not apply to the noise requirements of 3.6.4 and 3.6.5 nor shall it apply to sensing units unless specifically noted otherwise.

3.6.11 Electromagnetic interference. The recording set shall meet the electromagnetic interference requirements of MIL-STD-826 and the transient susceptibility test specified in 4.6.16.

3.6.12 Warmup. Warmup time for the recording set shall not exceed 5 seconds including automatic calibration sequence.

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3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. Consecutive serial numbers beginning with the number 1 shall be used. The procuring activity shall specify the numbering to be used for subsequent contracts.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Details of components

3.9.1 Recorder mechanism

3.9.1.1 Design. The recorder mechanism shall provide for mounting and operating the magazine assembly for data collection and all electrical or mechanical fittings required for operation of the recording set shall be contained in or on the recorder mechanism. The recorder mechanism shall contain the drive motor and gearing, calibration circuitry, power supplies, analog-to-digital converter, signal conditioners, multiplexer and logic, timer and programmer, selection circuitry and other components required for proper functioning. These components or subsystems shall be designed and built as individual modules so that they may be removed or replaced from the recorder mechanism with minimum time and maintenance difficulty.

3.9.1.2 Signal conditioning. Signal conditioning circuits shall be provided to:

a. Accept each of the 24 input signals specified by the procuring activity

b. Low pass filter

c. Convert the input into a standardized signal, such as 0 to 5V dc. Input low-pass filters shall be provided having 20 db attenuation at twice the maximum frequency and rolloff of 30 db per octave thereafter except at 400 Hz the minimum shall be 40 db and from 4 kHz through 100 mHz the minimum shall be 60 db.

3.9.1.3 Multiplexing. Multiplexing circuits shall be provided to sequentially sample the 24 input channels in such a manner that the information can be recorded and retrieved with the listed frequency response.

3.9.1.4 Analog-to-digital (A-to-D) conversion. A-to-D conversion circuitry shall be provided to convert the standardized input signal into digitized binary code for recording on the tape in the selected format consistent with accuracy requirements for the recording set.

3.9.1.5 Recording. A recording or magnetic tape shall be made of each sample, or each sample change, of the 6 or 7 (as applicable) most significant bits defining each sample along with corresponding pulses for clocking and synchronization, if necessary, and the address and time code to define the sample.

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3.9.1.6 BITE. BITE shall be provided for use in the self-test cycle and in a manual mode. The self-test cycle shall isolate the normal inputs, supply a standardized signal in sequence to each of the 24 input channels (handled through the signal conditioner, multiplexer, and computer as a normal function) and tested at the input to the tape heads for an expected code. Also, the tape drive motor is tested for current, and a possible tape breakage is checked. If any are incorrect, the record is flagged as NO-GO. The manual mode shall be initiated by a switch on the recording set and shall provide an indication of NO-GO to the serviceman.

3.9.1.7 Construction. The recorder mechanism shall be of easily accessible, simple construction; it shall provide for positive and secure attachment of the magazine assembly. The method of attachment shall assure minimum wear to moving or coupled components and a minimum of lag or backlash during the initial motion of the drive mechanism.

3.9.1.8 Elapsed time indicator (ETI). An ETI shall be so mounted within the recorder mechanism that the face of the indicator will be visible on the front. This indicator shall operate continuously whenever power is applied to the recording set. The ETI shall indicate HOURS in increments of 1 hour and shall have a full-scale range of 999 hours. The ETI shall have no provisions for manual RESET for any portion of the 999 hours.

3.9.2 MTK-43/A24U-6 mount. The mount dimensions shall be as specified by the procuring activity.

3.9.3 Magazine assembly

3.9.3.1 Design and construction. The magazine assembly shall be that portion of the recording set that contains the recording medium, supply and takeup reels, record/playback heads, erase head, guide rollers, and associated equipment as required for proper operation. The magazine assembly shall have provisions to prevent damage to the recording medium when being detached from or installed on the recorder mechanism or signal data converter. The magazine assembly shall provide for easy, secure, and positive mounting to the recorder mechanism or signal data converter wih no special tools required for connecting, disconnecting, or removing the assembly. Bolts and accessories necessary to satisfy this requirement shall be of the captive type and shall not protrude from the exposed surfaces of the recording set. The magazine assembly shall contain no protruding elements such as drive gears and electrical fittings. It shall be so enclosed as to provide protection against penetration of dust and foreign particles when detached from the recorder mechanism or signal data converter.

3.9.3.1.1 The design and construction of the magazine assembly shall provide a visual indication of the number of hours of remaining record time. Remaining record time shall be monitored from the supply reel.

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3.9.3.1.2 The supply and takeup reels shall be constrained when the magazine assembly is not attached to the recorder mechanism so that in transit or with handling, tape slack will not occur.

3.9.3.2 ETI. An ETI shall be mounted in the magazine so that the face of the indicator is visible from the magazine-recorder interface side of the magazine. This indicator shall operate continuously whenever power is applied to the recording set. The ETI shall indicate HOURS in increments of 1 hour and shall have a full scale range of 999 hours. The ETI shall have no provisions for manual RESET during any portion of the 999 hours.

3.9.3.3 Automatic turnoff. The magazine assembly shall automatically turn off the recording set when the tape has completed its travel from one reel to the other. This feature shall also apply to the playback mode of operation in both directions of travel, but shall only turn off the tape drive of the signal data converter. The turnoff mechanism shall operate in such a manner as to preclude the possibility of damage, breakage, or the necessity of rethreading the ends of the tape.

3.9.3.4 Interchangeability. The magazine assembly shall be directly interchangeable with all other magazines procured under this specification.

3.9.3.5 Recording medium. The recording medium (tape) shall be compatible with the crash, reliability, longevity, and intended use of the recording set.

3.9.3.6 Record/playback heads. The record/playback heads shall be an integral part of the magazine assembly and shall meet the specified reliability requirements. The estimated life expectancy and reliability of the heads shall be analyzed by the manufacturer's design and test data (considering all factors of intended use) and the results reported to the procuring activity. The specific design of the heads shall be approved by the procuring activity.

3.9.3.7 Crash survival. The magazine assembly shall withstand high-intensity impacts and high temperatures (in that sequence) that can be anticipated in a crash environment of high-performance jet aircraft. Impact loads shall be half-sine wave shocks of 3-millisecond duration and 1,000g's peak amplitude with application in any axis; ambient temperature shall be 871°C with a 20-minute exposure over the entire top and side areas of the magazine. In addition to high-temperature metallic-magnetic tape, the magazine cover shall be constructed of impact- and fire-resistant fiberglass material. Crash survival is the ability to readout data previously recorded on the tape after the tape has been subjected to the crash environments.

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3.9.3.8 Information plate. A plate containing the following information shall be provided on the magazine assembly.

Date Installed_____ Date Removed_____

Recording Set Serial No._____

Aircraft Type_____ Serial No._____

Aircraft Station or Squadron_____

Remarks:_____

3.10 Tape transport case. A shipping container shall be furnished for each magazine assembly. The container shall be so designed that it may be used to ship the magazine assembly by mail or other common carrier without damage. Magnetic shielding and a shock-absorbent material shall line the inside of the case. The case shall contain a handle to facilitate transportation.

3.11 Dimensions. Dimensions of the recording set shall be as specified by the procuring activity. Consideration shall be given to minimization of space for removal of the magazine assembly.

3.12 Weight. The weight of the recording set shall not exceed 30 pounds.

3.13 Workmanship. The recording set, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges, accuracy of dimensions, marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring and riveting, alignment of parts, and tightness of assembly screws and bolts. Machining, assembly, and calibration techniques shall be compatible with the high precision required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of recording sets shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

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4.2.1 Test outline (see 6.3). The contractor shall prepare an outline of tests and procedures to be used including methods of recording test data. The outline shall include all tests, including those specified herein, necessary to determine compliance with the requirement specified herein.

4.3 Test conditions

4.3.1 Attitude. The recording set shall be tested in its normal attitude with the mounting holes on a horizontal surface except during specific preproduction tests. During these tests, one recording set shall be completely tested with the mounting holes on a horizontal surface; one shall be completely tested mounted on a vertical surface; and one shall be completely tested mounted on a vertical surface but rotated 90° around an axis perpendicular to the base.

4.3.2 Warmup time. Prior to testing, a 5-minute period from POWER-ON to CALIBRATE shall be allowed for warmup. Power to the drive, motors shall be applied during this 5-minute period and a zero-scale input signal shall be applied to all channels.

4.3.3 Standard atmospheric conditions. When the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg), at room temperature (approximately 25°C), and at a relative humidity of not more than 90 percent. When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.4 Test facilities. Test facilities shall be in accordance with MIL-STD-810.

4.3.5 Measurements. Measurements shall be made as specified in MIL-STD-810.

4.3.6 Sequence of environmental tests. The sequence of environmental tests shall be in accordance with MIL-STD-810.

4.3.7 Tapping. Tapping or vibration of sensing units prior to the measurement of an output value or during any phase of the testing shall not be allowed.

4.3.8 Adjustments. Adjustments shall not be made during testing.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of five recording sets representative of the production equipment. The samples shall be identified by the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. After completion of the preproduction tests, the contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.10).

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

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4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests

4.5.1.1 Recorder mechanism and magazine assembly. Each recorder mechanism and magazine assembly shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Linearity

c. Noise (p-p)

d. Frequency check.

4.5.2 Sampling plan A

4.5.2.1 Recorder mechanism and magazine assembly. One recorder mechanism and one magazine assembly selected at random from each 25 or less produced on the contract shall be subjected as a signal unit to the following tests as described under 4.6:

a. Examination of product

b. Linearity (all channels)

c. Noise (rms) (all channels)

d. Noise (p-p) (all channels)

e. Drift (all channels)

f. Frequency response (all channels)

g. System accuracy (all channels)

h. Voltage and frequency regulation

i. Vibration (all channels)

j. Remote switch cycling

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k. Transient voltage

l. Dielectric strength

m. Power consumption

n. Temperature-altitude (all channels. To be conducted only once on a total production run; procuring activity to specify the unit to be tested)

o. Electromagnetic interference (to be conducted only once on a total production run; procuring activity to specify the unit to be tested).

4.5.2.2 Sampling plan B

4.5.2.2.1 Recorder mechanism and magazine assembly. One recorder mechanism and one magazine assembly selected at random from each 10 or less produced on the contract shall be subjected to the following tests as described under 4.6:

a. Individual tests

b. Temperature.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure; but final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.2.4 Sampling plan C. Three recording sets shall be selected at random from the first 10 produced and 6 from each 100 produced thereafter and subjected to the reliability test specified in 4.6.19 through 4.6.19.3.

4.5.2.4.1 Acceptance or rejection. Acceptance or rejection shall be in accordance with MIL-STD-781, test plan I, except as specified in 4.6.19. When any test group is rejected, no equipment on hand or later produced shall be accepted until the extent and cause of failure are determined. Corrections made shall be approved by the procuring activity.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.6 Test methods

4.6.1 Examination of product. The equipment shall be closely examined for proper identification and markings, overall configuration and workmanship, and proper voltages on all output connectors. Continuity of the magazine wiring and recorder mechanism shall be tested.

4.6.2 System accuracy. System accuracy shall be determined by introducing an appropriately filtered signal from a programmed signal generator into the inputs of all channels of the signal data recorder for 1 hour and comparing the reproduced output from the signal data converter to the filtered input. The filters shall have the same attenuation and bandwidth characteristics as the low pass filters supplied as part of the transducers, as applicable.

4.6.3 Linearity. Eleven equally spaced voltage levels from zero to full scale shall be recorded on all data channels except as specified otherwise. Each level shall be of 1-minute duration except as noted. The recorded data will be reproduced by the signal data converter and each of the 11 reproduced levels on each channel compared to the ideal output. The difference shall be considered error and the maximum error on any channel shall be considered the linearity error of the recorder. The recorder shall not be turned off between the 11 input steps.

4.6.3.1 The eleven ideal output levels shall lie equally spaced on a straight line drawn between the signal data converter ideal output end points. The ideal output end points shall be minimum voltage output for a zero level input signal and maximum voltage output for a full-scale level input signal.

4.6.3.2 The zero-scale and full-scale voltage input levels for each of the 24 channels will be stated by the procuring activity.

4.6.4 Noise (rms and peak-to-peak). The value of the peak-to-peak noise, including quantization noise and all other sources of noise, shall be checked on all steps of all linearity tests and the resulting values measured and recorded. Any noise pulse of a half sine wave nature shall be counted as a single pulse. The 10-minute period and corresponding distribution of noise pulses stated in 3.6.5 shall apply.

4.6.4.1 Both peak-to-peak and rms noise shall also be measured from the drift test specified in 4.6.5. A suitable meter shall be used for the rms measurements (Hewlett Packard 3400 A, or equivalent).

4.6.4.2 In the event one or more channels appear to be marginal as to meeting the noise specification, the procuring activity may require additional tests which shall not exceed 1 hour of recording.

4.6.5 Drift. A half-scale signal shall be applied to all channels of the recorder and held constant for an 8-hour recording period. The maximum deviation of the output from the original value shall be compared to full scale and the percent error

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calculated. Measurements of the reproduced signal shall be made at least every half-hour of recording time on all channels.

4.6.6 Frequency response. Frequencies encompassing the information bandwidth as specified in 3.5 shall be introduced into the appropriate channels of the recording set except 1 and 2. The peak-to-peak amplitude of this signal shall be 90 percent of the recording set ideal input range. Frequency response error shall be determined from this data after it is reproduced by the signal data converter. The peak-to-peak level at each of the frequencies listed below shall be compared to the D.C. level and the difference used for determination of percent error. The maximum error of any channel shall be taken as the frequency response error of the recording set. The format shown on figure 1 and the following step inputs shall be utilized:

Percent maximum bandwidth (per channel) 20, 40, 60, 80, 100.

4.6.6.1 Distortion. System distortion shall be determined by introducing an appropriately filtered signal from the programmed signal generator into the inputs of all channels of the signal data recorder for 1 hour and comparing the reproduced output from the signal data converter to the filtered input. The data shall have the same output characteristics and bandwidth characteristics as the input. Deviations from this input shall be considered as distortion. The distortion, when separated from the quantization error, shall be less than 1 percent.

4.6.7 Temperature sensitivity. The linearity test specified in 4.6.3 and the frequency response test specified in 4.6.6 shall be performed on the recorder at room temperature, +55°, +71°, -30°, and -54°C in the order listed. The recorder shall be stabilized at the specified temperature and then soaked for 2 hours with no power applied. After a 5-minute warmup period, the linearity test shall be conducted immediately followed by the frequency response test without restabilization of the recorder temperature. The linearity steps shall be of 1-minute duration. Power shall be turned off and on at least four times during the 5-minute warmup period.

4.6.8 Voltage and frequency regulation. The recorder shall be tested for proper operation and performance under the conditions listed in table I. The recorder shall be operated for 15 minutes under the specified voltage condition at the end of which time, while still at the same voltage condition, the test specified in 4.6.3 shall be performed. All data channels shall be played back through the signal data converter and checked for obvious failures and analyzed for linearity error. This test shall be performed at room temperature, +71°C, and -54°C.

TABLE I. Voltage Regulation Conditions

Condition	Frequency	Voltage
a	380 Hz - single phase	122V ac; 29V dc
b	420 Hz - single phase	105V ac; 17V dc
c	380 Hz - single phase	105V ac; 17V dc
d	420 Hz - single phase	122V ac; 29V dc

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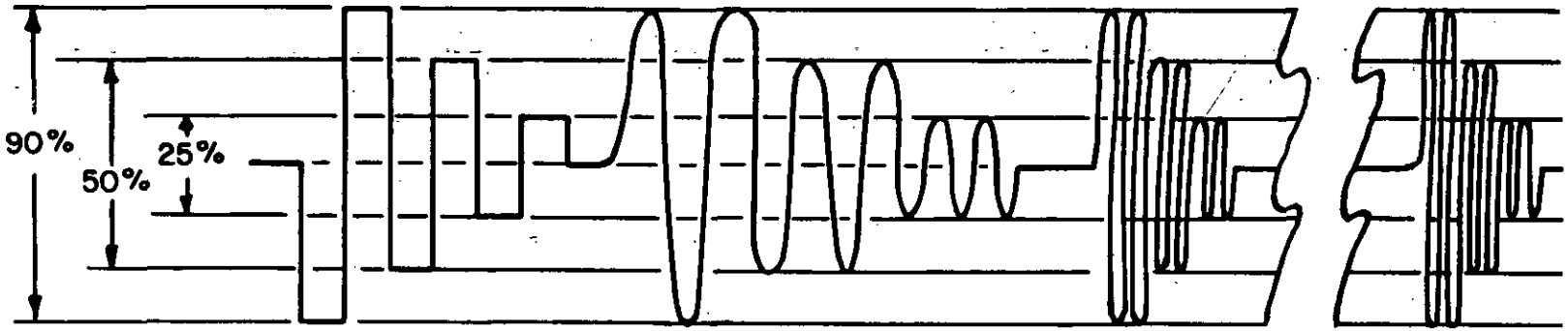


FIGURE 1. Programmable Signal Generator

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4.6.8.1 Remote switch cycling test. The recording set shall be tested for system malfunctions that may occur due to the cycling of the remote switch in an on/off mode. This test shall consist of simulations of switching at four frequencies. The following spectrum shall be used:

Frequency (Hz)	Duration (minutes)
5	4
2	4
4	4
6	4

This spectrum shall be conducted at room temperature, +71° and -54°C. At the completion of each 4-minute run, the recording set shall continue to operate in a remote switch on mode for 2 minutes. The magnetic tape shall be analyzed for system malfunctions.

4.6.8.2 Transient voltage. While operating from normal power, the recording set shall be subjected to three applications of voltage transients for a 10-minute (minimum) recording period. The three applications shall be as follows:

a. AC: 190V rms for 0.1 second
155V rms for 1.0 second
137V rms for 5.0 seconds

b. DC: 80V for 0.04 second
68V for 0.13 second
36V for 5.0 seconds.

The manufacturer shall select critical test points for testing in accordance with MIL-STD-704, limit 3 of figures entitled "Transient a-c voltage step function loci limits" and "Transient d-c voltage step function loci limits". Following application of the transients, the recording set shall be analyzed for noise and shall meet the requirements of 3.6.4 and 3.6.5. No damage to the equipment shall result from this test.

4.6.8.3 Dielectric strength. The recording set components shall have a potential of 500V rms at commercial frequency applied between the terminals of isolated circuits and between terminals of ungrounded circuits and the case ground for a period of 1 minute. The tester shall be set to operate between 0.5 milliampere and 10.0 milliamperes. There shall be no evidence of insulation breakdown. The manufacturer shall submit information relative to component circuits for which alternative tests are appropriate, or for which dielectric testing is not practical. Such circuits shall be subject to the approval of the procuring activity.

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4.6.8.4 Power consumption. The power required to operate the recording set shall be measured and shall not exceed the value specified in 3.5.4.

4.6.9 Preshipment testing. A continuous test of 24-hours duration shall be performed on each component of the recording set. Operation and data reduction of the recorder and the three sensors will be required.

4.6.9.1 Recording set. A continuous test of 24-hours duration shall be performed on the recording set. Operation of the total system, including all simulated transducers, will be required during the entire testing period. The simulated sensing units shall be in the normal static condition. One-hundred-foot cables of the type normally used in aircraft wiring for sensing units shall be used to connect the simulated transducers to the recording set. The recorded signals after playback through the converter shall be checked for obvious failures. The data shall not be erased. At the conclusion of the tests, with the equipment still operating, the following procedure shall be utilized:

a. The data from the last hour of the 24-hour test shall be analyzed for noise using the noise requirements of 3.6.4 and 3.6.5.

b. Simulated inputs shall be programmed into all 24 channels of the recording set. The inputs shall consist of three equally spaced voltage levels from zero to full scale in 30-minute steps. The data shall be recorded, processed through the converter, and measured for error. The maximum error shall not exceed the combined error allowable for the particular sensor and the recording set. This data shall not be erased.

c. A full-scale 10 percent of maximum bandwidth sine wave of at least 2-minutes duration shall be applied to all data channels. It shall be played back through the converter and checked for obvious failures. It shall not be erased.

4.6.9.2 Test plans. Test plans for transducers shall be submitted to the procuring activity for approval, as applicable. The test plans shall be similar to that specified for the recording set.

4.6.10 Temperature-altitude test. The recording set shall be tested in accordance with MIL-STD-810, method 504, equipment class 2 except 80,000 feet shall be used instead of 70,000 feet and 71°C shall be used in lieu of 95°C. A linearity format shall be used for performance tests. Temperature sensitivity tests (see 4.6.7) need not be performed on the recording set subjected to this test.

4.6.11 Vibration. The recording set shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, equipment category (b), curve H. This test shall be conducted at room temperature, +55°, and -30°C. The procuring activity shall choose the recorder speed mode for each vibration test. Inputs to the recorder shall be as follows:

a. Channels 1 through 12: Linearity format of 4.6.3

b. Channels 13 through 24: Full scale 10 percent of maximum bandwidth sine wave.

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4.6.11.1 Performance of the recording set during the vibration tests shall be within all tolerances specified herein. Noise levels (including jitter, dropins and dropouts, and other similar phenomena) shall be carefully analyzed along with linearity for adherence to the specified requirements. All indications of possible vibration susceptibility shall be immediately brought to the attention of the procuring activity.

4.6.11.2 All data channels will be played back through the converter and re-recorded on oscillographic paper at 10 inches per second. A count of the dropins and dropouts and their amplitude shall be made on the data from all channels.

4.6.11.3 Following the vibration tests, a room temperature linearity test shall be performed.

4.6.12 Shock, humidity, salt fog, and sand and dust tests. The recording set shall be subjected to these tests in accordance with the applicable methods of MIL-STD-810. After completion of each of the tests, the linearity format of 4.6.3 (except that 2-minute steps may be used) shall be introduced into all channels of the recording set and calibration tests shall be performed on the sensors. The data shall be analyzed for compliance with this specification.

4.6.13 Fungus and explosive atmosphere tests. The recording set shall be subjected to fungus and explosive atmosphere tests in accordance with MIL-STD-810, methods 508 and 511, procedure I, respectively.

4.6.14 Temperature shock. The recording set shall be subjected to a temperature shock test in accordance with method 503 of MIL-STD-810.

4.6.15 Acceleration. The recording set shall be subjected to an acceleration test in accordance with method 513 of MIL-STD-810.

4.6.16 Electromagnetic interference control. The recording set shall be subjected to an electromagnetic interference control test in accordance with MIL-STD-826 for class AM, type E equipment. All required and optional tests shall be conducted.

4.6.16.1 Transient susceptibility. The recording set shall operate without malfunction or degradation of performance when exposed to a switching transient field generated as shown on figures 2 and 3. The lead length between the recording set and dummy transducers shall be 70 feet. The dummy transducers shall be passive networks of resistors, capacitors, and chokes as required to simulate the passive characteristics of the transducers. The relay shall be in accordance with MS25271, or equivalent, and shall be wired for continuous operation. No suppression shall be applied to the relay. The relay circuit shall be unshielded wire, shall be taped to and paralled with the equipment input power leads, signal leads, and interconnecting leads, and shall be looped about the black boxes of the test set.

4.6.17 Frequency check. A 90-percent full-scale signal at the minimum bandwidth specified in 3.5 shall be introduced into the appropriate channels of the recording set

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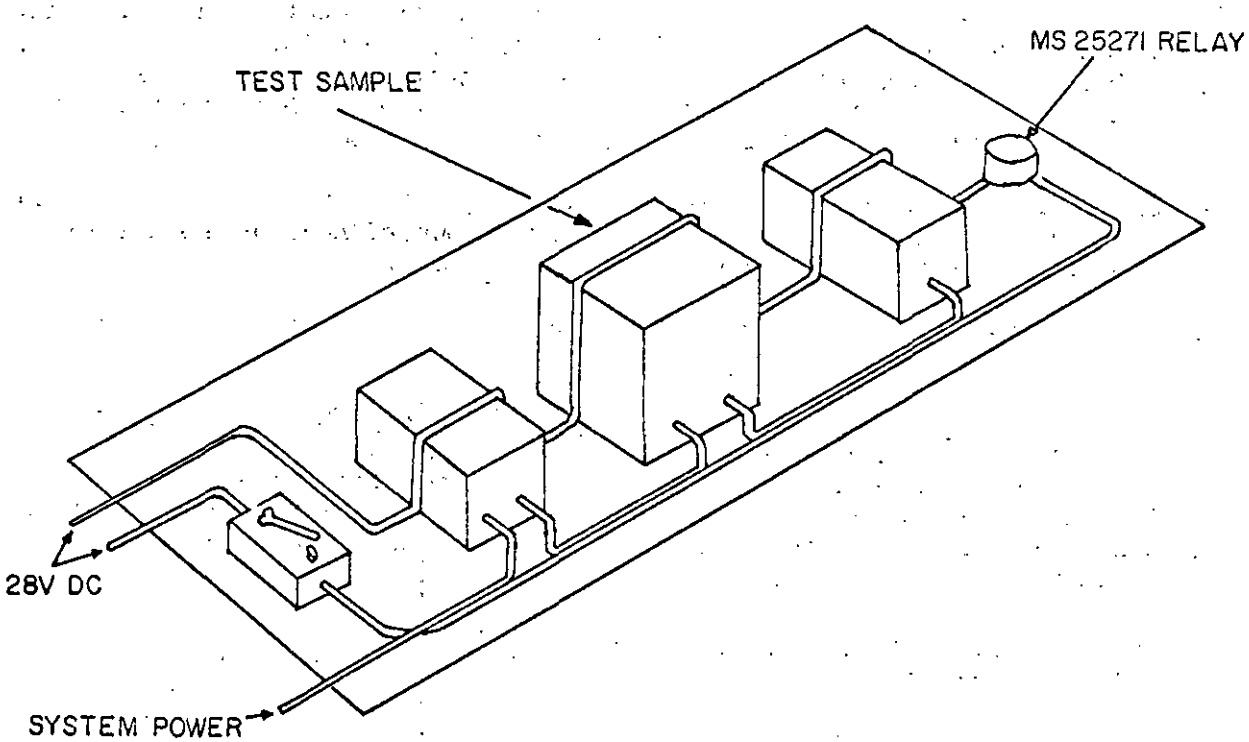


FIGURE 2. Transient Susceptibility Test

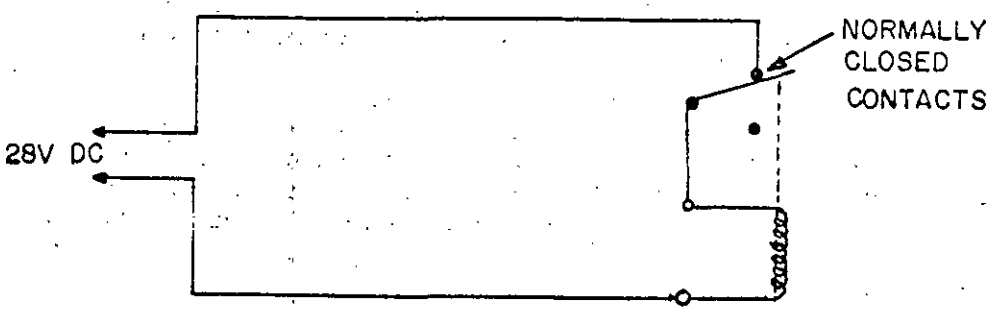


FIGURE 3. Relay Wiring Details

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for at least 30 seconds. This data shall be played back through the converter. The peak-to-peak value on playback shall be within 2.0 percent of the ideal value.

4.6.18 Digital output. A linearity format of 11 equal steps of 60 seconds duration per step shall be introduced into each channel of the recording set for 30 minutes duration. This shall be reproduced onto digital tape. The digital tape shall be printed out, and the entire data shall be carefully analyzed for exceedance of the linearity, noise level, and other requirements of this specification. A carbon copy of the printout and detailed records of the results shall be furnished to the procuring activity.

4.6.18.1 All digital control functions shall be checked for proper operation.

4.6.19 Reliability. The recorder shall be subjected to a reliability test in accordance with MIL-STD-781, test level F, and as specified herein, which shall demonstrate a specified MTBF of 400 hours. Each of the three recorders shall be tested a minimum of 800 hours for a total minimum test time of 2,400 hours, or six times the MTBF, before an accept decision is reached. The thermal survey specified in MIL-STD-781 shall also apply.

4.6.19.1 Performance check. A reliability performance check shall be accomplished using simulated sensor inputs and shall consist of programming into all 24 channels at least three equally spaced voltage levels from zero to full scale in 30-second steps. The data shall be played back through the signal data converter and recorded for data analysis as required to adequately determine compliance with this specification. A full test procedure shall be developed by the contractor for approval by the procuring activity; it shall include noise measurements and checks on the accuracy of the above signals. Included also shall be provisions to operate and cycle the sensing units during all reliability testing. An 11-step linearity test utilizing simulated inputs format shall also be used for the performance test.

4.6.19.2 Recording periods (minimum). The recording periods shall be not less than 8 continuous hours with the power ON, motor running, and the magazine properly operating.

4.6.19.3 Longevity. The longevity test shall be conducted in accordance with MIL-STD-781, test level F, for two predicted MTBF periods on the three test samples specified in 4.6.19. The test level shall then be changed to A1 of MIL-STD-781 for an accumulation of 4,500 hours on one unit which shall include the test time specified in 4.6.19. During testing at test level A1, data shall be analyzed every 100 hours. Data covering servicing and parts requiring replacement in less than 10,000 hours shall be submitted to the procuring activity for review and approval.

4.6.20 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471, appendix A, method 4.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The A/A24U-6 recording set covered by this specification is intended for used in the collection and processing of structural loads encountered by weapon systems and flight vehicles. The recording set playback measures and records time histories of the principal flight parameters from which structural loads experienced by weapons systems in tactical roles can be detected. The collected data will be analyzed by an automatic data processing facility in order to determine the structural loads actually encountered during tactical maneuvers of flight vehicles and weapon systems. Appropriate processing of data will also establish fatigue limits and modification requirements for present aircraft, prescribe design parameters for new systems, and accurately project IRAN requirements.

6.2 Ordering data. Procurement documents should specifiy the following:

a. Title, number, and date of this specification

b. Mounting configuration

c. Number of spare magazines to be furnished

d. Number of sensing units to be furnished

e. Sensing unit ranges and frequency response

f. Sensor electrical characteristics

g. When design data is required (see 6.4)

h. Required levels of preservation, packaging and packing.

6.3 Test outline. The test outline (see 4.2.1) will be submitted, in triplicate, in workbook report form allowing space for recording data, sketches, and notes. The test outline should be submitted to the procuring activity for approval 30 days prior to formal testing. If, during the test program, the procuring activity finds the test outline to be inadequate, it will be amended as necessary.

6.4 Approved design. The design used in the fabrication of the recording set shall be approved by the procuring activity. When specified (see 6.2) the contractor will prepare design data to be submitted with or prior to the test outline specified in 6.3.

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6.4.1 Design data. When available, the contractor will furnish the required data as prepared in the normal performance of the contract in lieu of the following data:

a. Brief operating instructions

b. A complete schematic diagram reduced to its simplest form showing circuits of all assemblies and subassemblies. Components not internal therein will be shown individually in schematic form with electrical interconnections indicated

c. A practical wiring diagram of each assembly, or of each component thereof, showing the physical location and connections of detail parts and subassemblies with reference symbols and terminal numbers indicated

d. A cabling diagram of the complete recording set

e. Outline dimensional drawings of all major and minor components and assemblies

f. Overall functional block diagram.

6.5 Vibration isolators. Mounting and space allocation for the recording set may vary among service aircraft; mounting orientation, total space available, and working clearances are factors. The procurement document may specify vibration isolators of a configuration different from that specified in 3.9.2; however, the recording set will be subjected to the vibration test specified in MIL-STD-810 at least once if the configuration is changed.

6.6 Spare magazine assembly. The procurement document may specify a number of spare magazine assemblies to be furnished under the contract. Of all magazine assemblies furnished, the procuring activity will designate, at random, which magazines are to be designated spares and which magazines are to be combined with recorder mechanisms and designated recording sets. Testing for acceptance of the spare magazines will consist of the individual tests specified in 4.5.1.1.

6.7 Sensing unit ranges. The recording set may be used in different types of service aircraft with widely varying performance capabilities. The procurement document should specify the range and frequency response of the sensing units to be furnished; other requirements and characteristics of the sensing units will remain the same.

6.8 Tape rewind. Consideration to moving the magnetic tape away from the record/reproduce heads during the rewind mode of operation should be given in the design of the magazine assembly. This feature would extend the life of the record/reproduce heads.

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6.9 Test report (see 4.4.2). The contractor will submit three complete copies of the test report to the procuring activity.

6.10 Marginal indicia. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activity:
Air Force - 11, 71

Project No. 6610-F033

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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